Feasibility of Machine Learning for Credit Default Prediction: An Experimental Report

1. Business Context

The start-up under consideration operates in the financial technology (FinTech) sector, providing consumer credit products to individuals who may be underserved by traditional banking institutions. As described in the Unit 9 report, credit risk assessment is a high-priority use case for deploying artificial intelligence within this organisation. The core business problem is predicting, at the point of application, whether a customer will default on their credit obligations. Accurate prediction enables the company to approve creditworthy applicants and limit losses from non-repayment. Manual underwriting processes are currently slow and inconsistent, relying on a limited set of financial indicators. An automated machine learning solution could deliver faster, more consistent, and data-driven decisions. However, as the Unit 9 report noted, any predictive model used in lending must be interpretable: regulators and affected customers have a right to understand how credit decisions are made (European Parliament and Council of the European Union, 2024). This experiment was designed to test the technical feasibility of such a system.

2. Justification of the Choice of Dataset

The dataset used is the Default of Credit Card Clients from the UCI Machine Learning Repository (Yeh and Lien, 2009), comprising 30,000 records of credit card holders with 23 features: credit limit, demographics (age, sex, education, marital status), six months of repayment history, monthly bill amounts, and previous payment amounts. The target variable is binary, indicating whether a client defaulted the following month. This dataset was selected for three reasons. First, its structure mirrors data held by a FinTech lender, allowing findings to transfer directly to the business context. Second, it is complete, with no missing values, and large enough for robust model training and testing. Third, its wide use in machine learning research provides a basis for contextualising the results obtained (Yeh and Lien, 2009). The dataset is publicly available from the UCI Machine Learning Repository (UCI Machine Learning Repository, 2016).

3. Justification of the Approach to Developing the Prediction Model

The experiment used a standard supervised machine learning process (where labelled data trains the algorithm to make predictions) for binary classification on tabular data, meaning the model predicts one of two possible outcomes from structured, row-and-column data. After loading and inspecting the data, an

imbalance in the target variable was noted: 23,364 non-default cases (77.9%) and 6,636 default cases (22.1%). This imbalance reflects real-world credit data and is likely to suppress model performance on the minority class (the less common outcome, here ‘default’).

Feature preparation was kept minimal, consistent with the traditional machine learning approach outlined in the Unit 9 report. Human-engineered features, created using domain knowledge rather than raw data, served as the primary inputs to statistical models (Theobald, 2024). All 23 features were retained; only the non-informative customer ID column was removed. Data were split using a stratified random split—80% for training (n = 24,000) and 20% for testing (n = 6,000)—with a fixed random seed to ensure reproducibility. Standard scaling was applied to normalise feature values for logistic regression, setting each feature to a mean of 0 and a standard deviation of 1. Tree-based models were trained on unscaled data, as decision trees and random forests are scale-invariant and unaffected by the numerical range of inputs.

The train-test split was chosen to give a reliable measure of how well each model generalises to new, unseen data—the critical question for a business considering deployment. All code was written in Python using scikit-learn within a Jupyter Notebook, ensuring the full pipeline is transparent, reproducible, and auditable.

4. Rationale for the Machine Learning Algorithms Used

Three algorithms were selected to enable comparison across the interpretability-to-accuracy spectrum outlined in the Unit 9 report: logistic regression, decision trees, and random forests.

Logistic regression served as the baseline and most interpretable model. It estimates the probability that a case belongs to a class by combining input features in a weighted sum. The coefficients show the direction (positive or negative effect) and strength (magnitude) of each feature’s impact on the prediction (Russell and Norvig, 2021). In regulated lending, this transparency is valuable: the model’s outputs can support specific, auditable explanations for credit decisions that meet EU AI Act requirements (European Parliament and Council of the European Union, 2024). Logistic regression is also computationally efficient and well-established in credit scoring applications (Veloso et al., 2021).

A decision tree was chosen as an intermediate option. It maintains interpretability—the logic behind each prediction can be followed step by step—while also handling non-linear relationships between features. Decision trees partition the feature space using threshold-based rules that can be visualised as a diagram, making individual predictions traceable (Theobald, 2024). The maximum tree depth was limited to five levels to reduce overfitting and preserve explainability.

Random forests were selected for their strong predictive performance as ensemble methods. They combine predictions from 100 decision trees, each trained on a random subset of the data, reducing variance and typically achieving greater accuracy than a single tree (Theobald, 2024). The increased accuracy comes at a cost: individual predictions cannot be traced through a single decision path, limiting the model's suitability for regulated contexts that require explanations. Feature importance scores offer only partial insight and do not constitute full explainability.

5. Analysis of the Outputs with Evidence of Testing

Table 1 summarises how all three models performed when tested on the reserved set of 6,000 examples not seen during training.

Model	Accuracy	Default Recall	Default F1	Interpretability
Logistic Regression	0.810	0.24	0.35	High
Decision Tree	0.819	0.37	0.47	Medium
Random Forest	0.816	0.37	0.47	Low

Table 1: Comparative model performance on the test set (n = 6,000).

All three models achieve a similar overall accuracy of 81–82%. However, overall accuracy can be misleading when classes are imbalanced: a model that predicted no defaults for any customer would reach 77.9% accuracy while offering no real predictive power. More informative measures are recall and F1-score for the Default class, which indicate how well the model identifies actual defaulters.

On these measures, logistic regression performs substantially worse than the tree-based models. It correctly identifies only 24% of actual defaulters in the test set. Each decision tree and random forest achieves a recall of 37% for the Default class, correctly identifying 483 of 1,313 defaulters. The confusion matrices (Figure 1) illustrate this clearly: logistic regression missed 1,002 defaulters, while the tree-based models each missed 830.

From a business perspective, missed defaulters represent false negatives: the model approves credit for customers who subsequently fail to repay, resulting in direct financial loss. False positives, creditworthy customers incorrectly declined, carry a lower financial cost but create reputational risk. All three models are biased towards the majority class due to the dataset's imbalance, producing a high false-negative rate. This systematic bias is the primary performance limitation identified by the experiment and must be corrected before deployment.

The feature importance analysis from the random forest model (Figure 2) shows that PAY_0, the most recent repayment status, is the strongest predictor, with an importance score of approximately 0.10. Age, recent bill amounts, and credit limit also rank highly, consistent with domain knowledge that recent payment behaviour is the strongest near-term signal of default risk. The concentration of predictive power in a small number of payment-related features suggests that a simpler, reduced-feature model could achieve comparable performance, which is a practical consideration for production deployment.

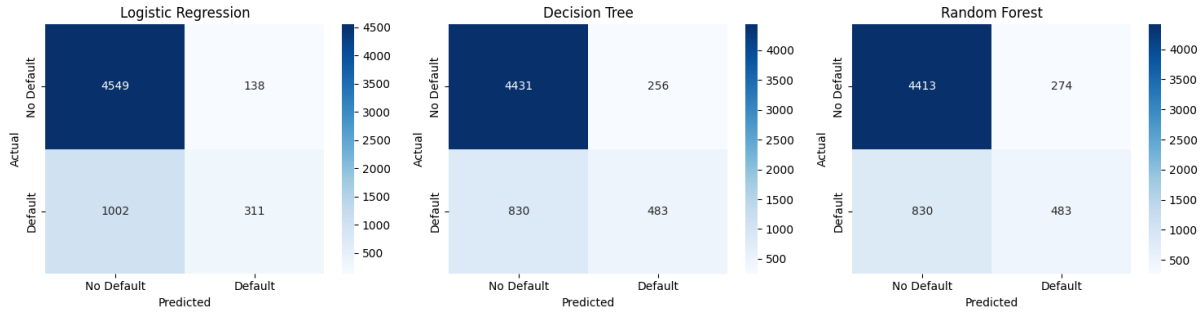


Figure 1: Confusion Matrices

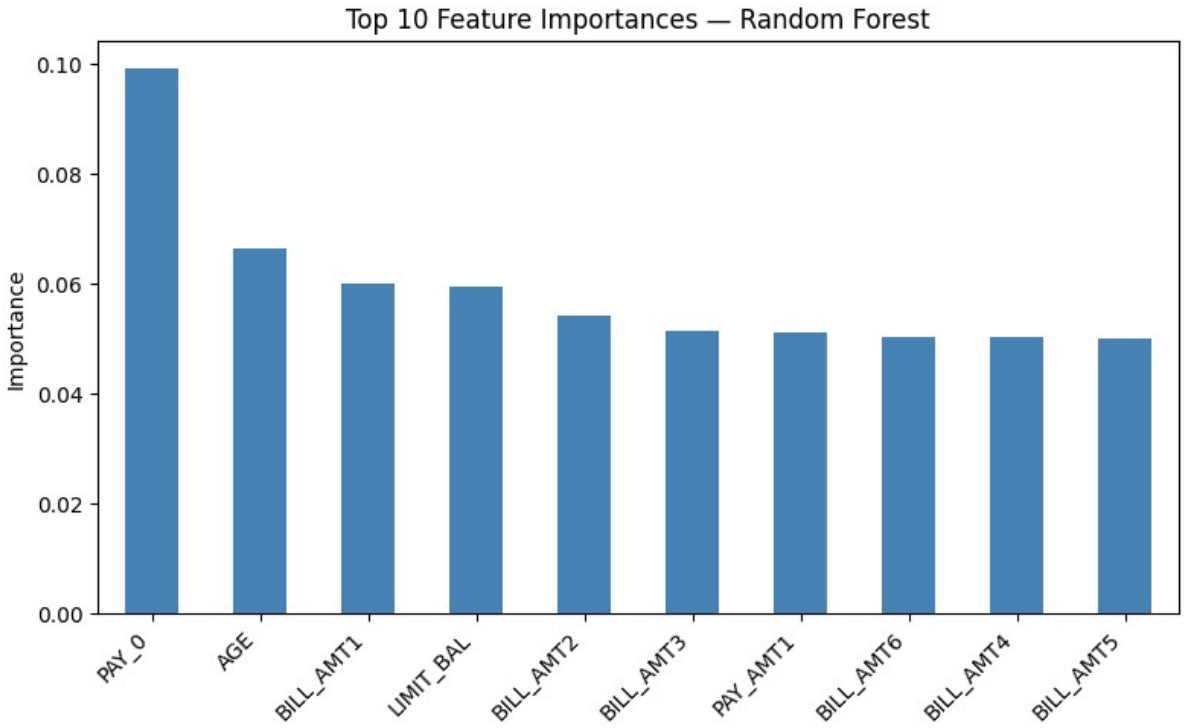


Figure 2: Feature Importance Chart

6. Demonstration of Transferability to the Unit 9 Use Case

The Unit 9 report argued that AI investment should align with the organisation's data, regulations, and risk appetite, rather than being driven by novelty. The results of this experiment provide empirical support for that argument and demonstrate how the pipeline could be implemented within the start-up.

The experimental pipeline—data ingestion, feature preparation, model training, and evaluation—is directly transferable to the start-up's operational environment. In place of the UCI dataset, the company would use its own customer records, which are expected to share the same structural characteristics: demographic data, credit history, repayment behaviour, and known default outcomes for historical cases. The Python code developed in this experiment could ingest these proprietary records with minimal modification, and the same evaluation methodology—training on historical data and testing on a held-out set—would apply without change.

The experiment also provides empirical support for the Unit 9 report's central argument regarding interpretability. Logistic regression achieves overall accuracy within two percentage points of the random forest model, yet offers substantially greater transparency. In a lending context, where the EU AI Act and consumer credit regulations may require the organisation to provide a specific, justifiable reason for an adverse credit decision (European Parliament and Council of the European Union, 2024), this transparency is not merely desirable but potentially a legal requirement. The experiment demonstrates that a logistic regression model can be deployed to meet this obligation, accepting a modest reduction in default recall relative to the ensemble approach. This trade-off mirrors the argument advanced in the Unit 9 report regarding credit application rejection: the more accurate model is not necessarily the better investment when the cost of opacity outweighs the marginal gain in predictive performance.

The experiment also identifies a limitation that needs to be addressed before production deployment: low recall for the Default class across all models, attributable to class imbalance in the training data. Techniques such as class-weight adjustment during model training, oversampling the minority class using methods such as SMOTE, or adjusting the classification threshold to increase sensitivity could be applied to improve default detection. These extensions are beyond the scope of this feasibility study but represent the immediate next steps in a development roadmap. The findings therefore confirm not only that the technology is feasible, but that the path from experimental prototype to a production-ready system is well-defined and achievable with widely available tools and expertise.

7. Conclusion

This experiment demonstrated the feasibility of using traditional machine learning algorithms to predict credit card default, directly addressing the use case identified in the Unit 9 report. Three models were trained and evaluated on a publicly available dataset analogous to the start-up's customer data. Decision trees and random forests achieved the best balance between accuracy and default recall, while logistic regression offered the greatest interpretability. The results confirm that a functional credit scoring model can be built using standard Python tools and that the pipeline is transferable to the company's own data environment. Class imbalance handling is identified as the primary recommended enhancement before production deployment.

References:

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